

MATH 448-548
Final project assignment

The projects are due December 11 (Thursday) at 11:59 p.m. The submission must be in the form of a single .pdf file containing all parts of your project.

The file name should be

LASTNAME_final.pdf

Preparation instructions:

- Write your full name and student ID number on the front page.
- Write all the codes from scratch. Do not use standard routines for ODE and PDE solving.
- The solution to every problem should contain three parts: your code (if required), the output (graphs and tables, as appropriate), and a written part where you do analysis and discuss the results. The code and output **MUST NOT** be handwritten. Screenshots of the actual MATLAB, Python, etc. code and computer-generated output only must be used.
- The written part should be either (i) typed using LATEX or some other word processor containing an equation editor, or (ii) written carefully by hand using a black pen.
- The file of your project should be submitted to Canvas.

Problem 1.

0. Preliminary information and data. In part 2, use the matrix

$$A = \begin{pmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \\ 1 & 3 & 3 \end{pmatrix}. \quad (1)$$

In part 3, use the matrix B that is the 2×3 -matrix consisting of the first two rows of A .

Singular value decomposition (SVD) and pseudo-inverse. Given a matrix $M_{m \times n}$ with $\text{rank}(M) = r$, write

$$M = U \Sigma V^T$$

where $U_{m \times m}$ and $V_{n \times n}$ are orthogonal matrices, and $\Sigma_{m \times n}$ is a diagonal matrix with nonzero diagonal entries $\sigma_1 \geq \sigma_2 \geq \dots, \sigma_r > 0$ called singular values of M .

Definition. Given M as above, the pseudo-inverse M^+ is defined by

$$M^+ = V \Sigma^+ U^T, \quad (2)$$

where $\Sigma_{n \times m}^+$ is the diagonal matrix whose non-zero entries are $\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \dots, \frac{1}{\sigma_r}$.

Thresholding operator.

The **hard thresholding operator** H_s is defined as follows.

$$\mathbf{x} \rightarrow H_s(\mathbf{x}) \in \mathbb{R}^n \quad (3)$$

$H_s(\mathbf{x})$ sets all entries of $\mathbf{x}$ to zero except for s largest (in absolute value) entries.

The matrix A_S . Given a matrix A and an index set S , denote by A_S the matrix consisting of columns of A with index in S .

Assignments and questions.

1. Implement the pseudocode for the Orthogonal Matching Pursuit (OMP). The pseudocode is contained in a separate handwritten note attached to the Final project assignment. You may use the standard Matlab (or other language) commands to compute SVD and pseudo-inverse.
2. Run the code for $\mathbf{b} = A\mathbf{e}_1$ with $\mathbf{e}_1 = (1, 0, 0)^T$. See how well the obtained vector $\mathbf{x}$ matches $\mathbf{e}_1$.
3. Run the code again, this time using the matrix B instead of A .
4. Compare the results obtained in parts 2, 3. Explain the connection with compressed sensing and sparse recovery.

Problem 2. Solve Computer problem 1 in Section 8.3. Write you own program from scratch. Then compare with the exact analytic solution that can be obtained by the method of integrating factors studied in Math 315. The exact solution is

$$x(t) = 12 \frac{e^t}{(e^t + 1)^2}.$$

Problem 3. Solve Computer problem 3 in Section 8.4.